1.530 Shane thinks of three integers. Added two at a time, their sums are 37, 41, and 44. What is the product of the three integers?

1.340 Keith wants to mark all of the original prices in his store down to half-off. The prices are already marked down by 20%. What additional percent should be discounted? Express your answer as a percent rounded to the tenth.

1.650 Congruent circles of radius 6 have their centers at C(5, 4) and D(1, 2). The circles intersect at points A and B. What is the slope-intercept form equation of the line which passes through A and B?

1.661 Find the perimeter of the smallest triangle whose integer side lengths form a geometric sequence whose common ratio r is not equal to 1.
(See **p.178**).

2.210 How many distinct pairs of consecutive integers have a product less than 10,000?

2.120 Ten friends go out to dinner together: 7 order an appetizer, 5 order a soup, and 4 order a salad. If everyone orders something, but no one orders exactly two things, how many people order all three things?

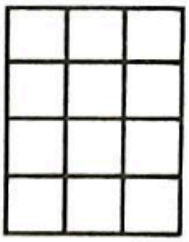
2.520 Students were surveyed on their preference for tea. Three-fourths of the students surveyed like sweet tea, three-fifths of the students surveyed like unsweetened tea, and one sixth of the students do not like either. What is the fewest number of students surveyed who could like both sweet and unsweetened tea?

2.811 How many pentagons can be formed by tracing the lines of the figure below?



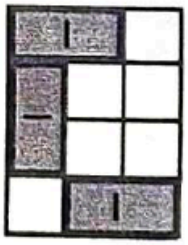
2.121 There are seven parking spaces in a row which must be assigned to four co-workers. How many ways can the spaces be assigned if at least two of the assigned spaces must be adjacent?

2.441 Three identical dominoes are placed on the 3×4 tiled board below to create a complete pathway from the upper-left corner to the lower-right corner of the board (no dominoes may touch only at the corners). How many ways can this be done?





Yes



No

2.461 Three friends: Al, Bill, and Carl, are painting their chests with the word CAVALIERS for a basketball game, with at least two letters painted on each friend's chest. For example, Al might get "CA", Bill "VAL", and Carl "IERS", or Carl could get "CAVA" while Bill gets "LI" and Al gets "ERS". How many ways can the three friends get their chests painted so that when they stand in the correct order, they spell-out the word CAVALIERS?

2.561 An elaborate code consists of left and right arrows which are placed together to represent words. There are spaces between words. For example, the code of three left and four right arrows arranged with two spaces: << >>> <> represents a three-word code phrase. How many different three-word code phrases in this system use 8 letters?

2.181 How many integers between 1,000,000 and 2,000,000 have at least four fives?

2.381 Three standard six-sided dice are rolled: one red, one green, and one white. How many outcomes are possible where the number rolled on the red die is greater than the number rolled on the green die, and the number on the green die is greater than the number rolled on the white die?